1.2.5 ▶ ▶ Reduced Mass Approach

In this section, we consider a method to approach a two-body system by reducing it to a singly body problem. Start by considering two bodies of masses m_1 and m_2 which interact via some interaction force F. The accelerations of each body are

$$a_1 = \frac{F}{m_1}$$

$$a_2 = -\frac{F}{m_2}$$

Where the signs are opposite because of Newton's third law. If we transform into the reference frame of m_1, m_2 experiences a force of magnitude F and accelerates at the relative velocity of the two bodies a_{rel}. Where

$$a_{rel} = \frac{F}{m_1} + \frac{F}{m_2}.$$

Rearranging gives

$$\begin{aligned} F &= \left(\frac{1}{m_1} + \frac{1}{m_2}\right)^{-1} a_{rel} \\ &= \mu a_{rel} \end{aligned} \tag{1.31}$$

where $\mu = \left(\frac{1}{m_1} + \frac{1}{m_2}\right)^{-1} = \frac{m_1 m_2}{m_1 + m_2}$ and is known as the reduced mass. In other words, if we transform into the reference frame of one of the bodies in a two-body system, the mass of the other body effectively becomes the reduced mass, and its motion is defined by Newton's second law ★